

EpiWatch

AI-Powered Multi-Disease Outbreak Prediction System for India

Real-time Outbreak Forecasting | 9 Districts | 3 Diseases | 54 Models

HistGradientBoosting + XGBoost Ensemble | IDSP | NASA POWER | Census 2011

The Problem

India's disease surveillance system (IDSP) produces weekly reports, but by the time outbreaks are detected through traditional reporting, they have already spread. There is no public, district-level early warning system that links climate triggers with historical case patterns to predict outbreaks BEFORE they peak.

Key Challenges

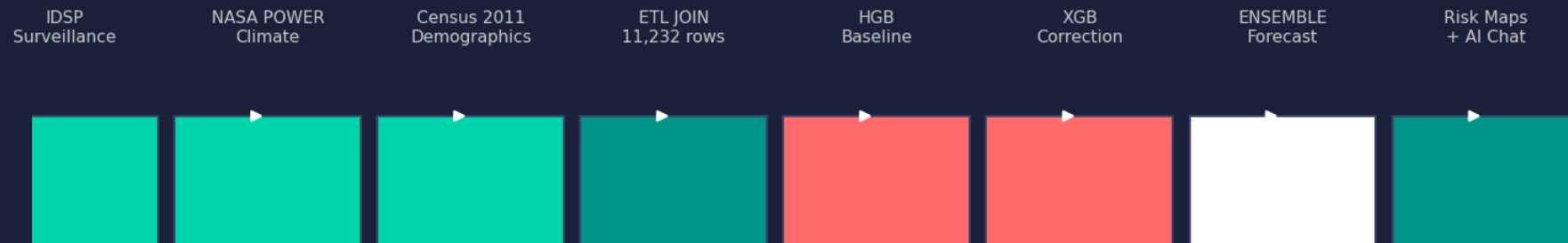
- Weekly reporting lags the actual outbreak peak
- No district-level early warning exists
- Climate - disease relationship is not modeled
- No real-time risk prediction available

Our Solution

- Real-time climate-data-driven forecasting
- District-level 8-week outbreak prediction
- Explainable AI with SHAP attributions
- Grounded AI assistant with source citations

The Idea

End-to-End Prediction Pipeline



EpiWatch closes this gap with a complete pipeline - from raw data ingestion to interactive risk maps - predicting outbreaks weeks in advance by learning the climate-disease relationship for each district.

ML Architecture

HistGradientBoostingRegressor

Seasonality Baseline

sin_week, cos_week features
cases_lag1, lag2, lag4
Captures seasonality & trend
max_iter=150, lr=0.08



XGBoost Regressor

Climate Residual Correction

rainfall_mm, rainfall_lag2
temp_max_c, temp_max_lag1
humidity_pct, humidity_lag1
n_estimators=80, depth=4

ENSEMBLE PREDICTION

Final cases = HGB baseline + XGB climate correction

8-week forward forecast with confidence intervals (0.8x - 1.25x) and risk tiering

Training Pipeline

For each of 9 districts x 3 diseases: train HGB -> compute residuals -> train XGB on residuals -> combine -> evaluate MAE/RMSE -> save .pkl

Result: 54 models (27 district-disease x 2 model types) | 0 training failures | avg MAE = 2.64

Data Sources & Feature Engineering

IDSP Surveillance

Weekly cases & deaths
9 districts · 3 diseases
104 weeks of data

NASA POWER API

Temperature, rainfall, humidity
Daily · 2023-2025
per district centroid

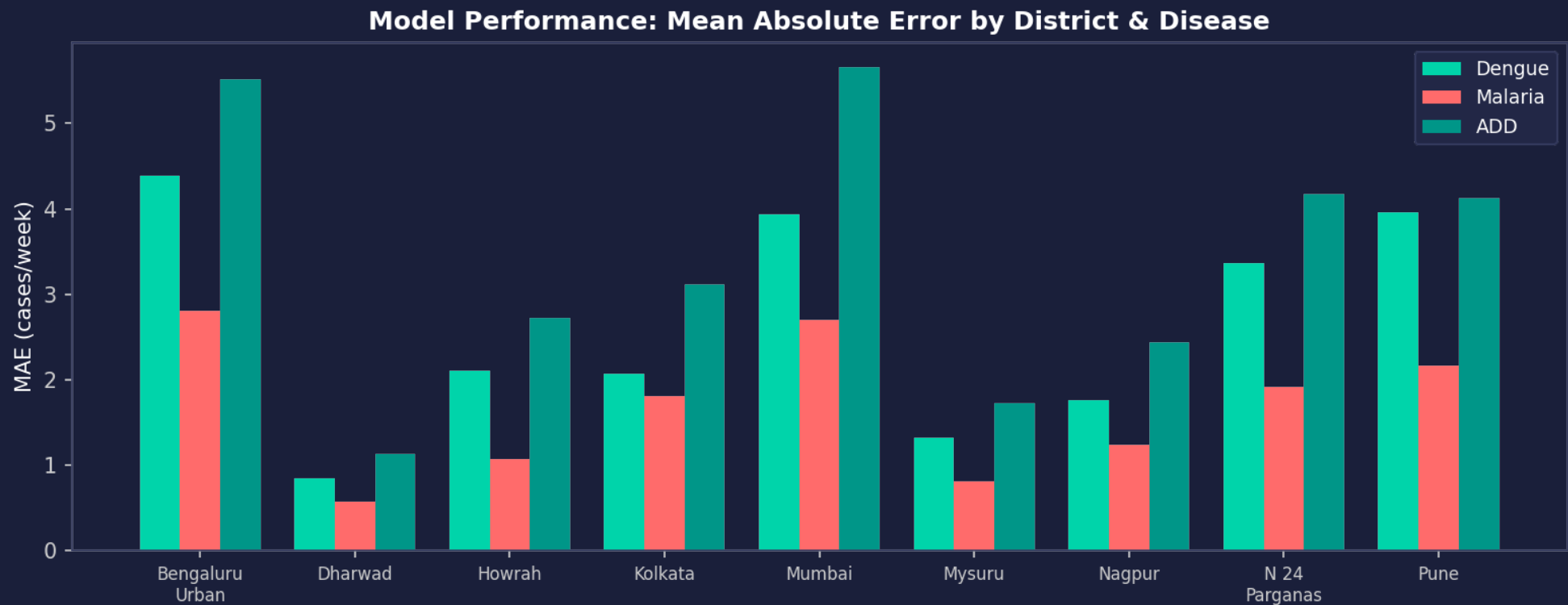
Census 2011

District populations + 2025
projections for per-capita
risk calculations

Feature Engineering

Seasonality	sin_week, cos_week (cyclical week-of-year encoding)
Case lagging	cases_lag1, lag2, lag4 (autoregressive history)
Rainfall	rainfall_mm (current), rainfall_lag2 (2-week delayed monsoon trigger)
Temperature	temp_max_c (current), temp_max_lag1 (heat spike driver)
Humidity	humidity_pct (current), humidity_lag1 (vector breeding conditions)

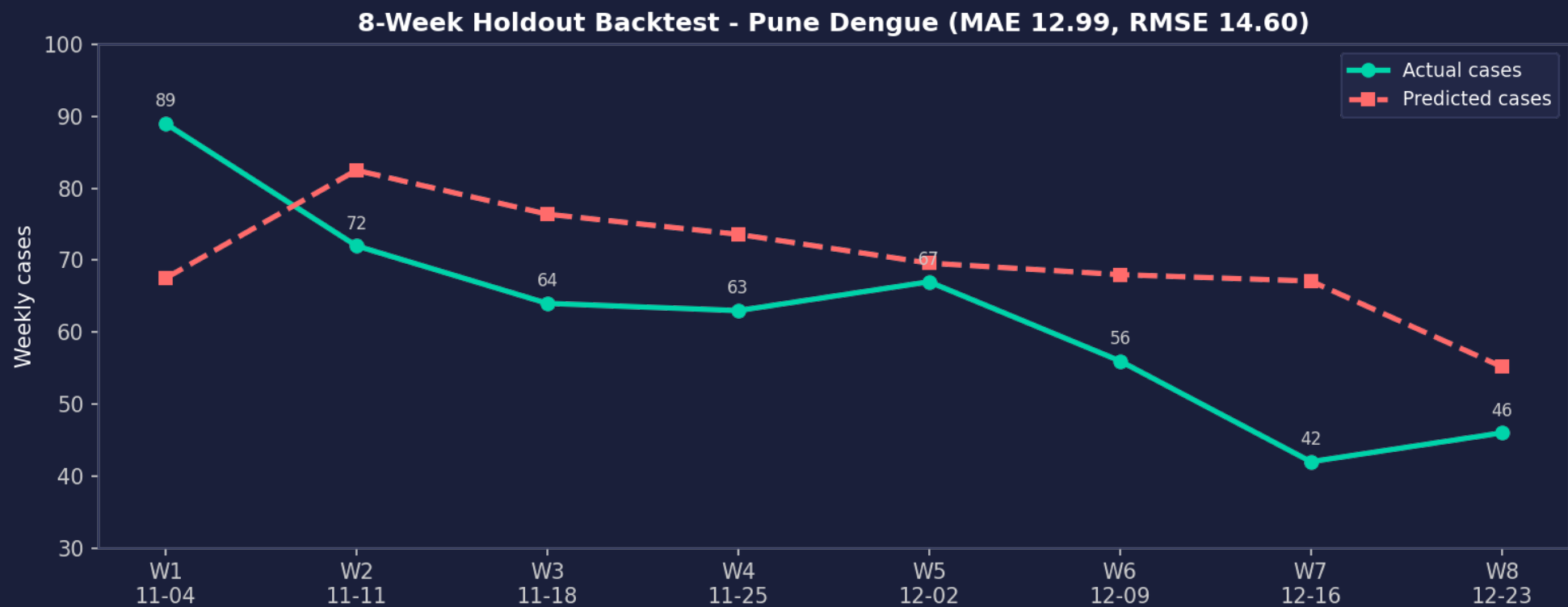
Model Performance



Cross-district average MAE: 2.64 | 27/27 district-disease models successful | 0 training failures

Worst: Mumbai ADD 5.66 | Best: Dharwad Malaria 0.57 | Backtest: MAE 12.99 / RMSE 14.60 (Pune Dengue)

Backtest Proof - 8-Week Holdout

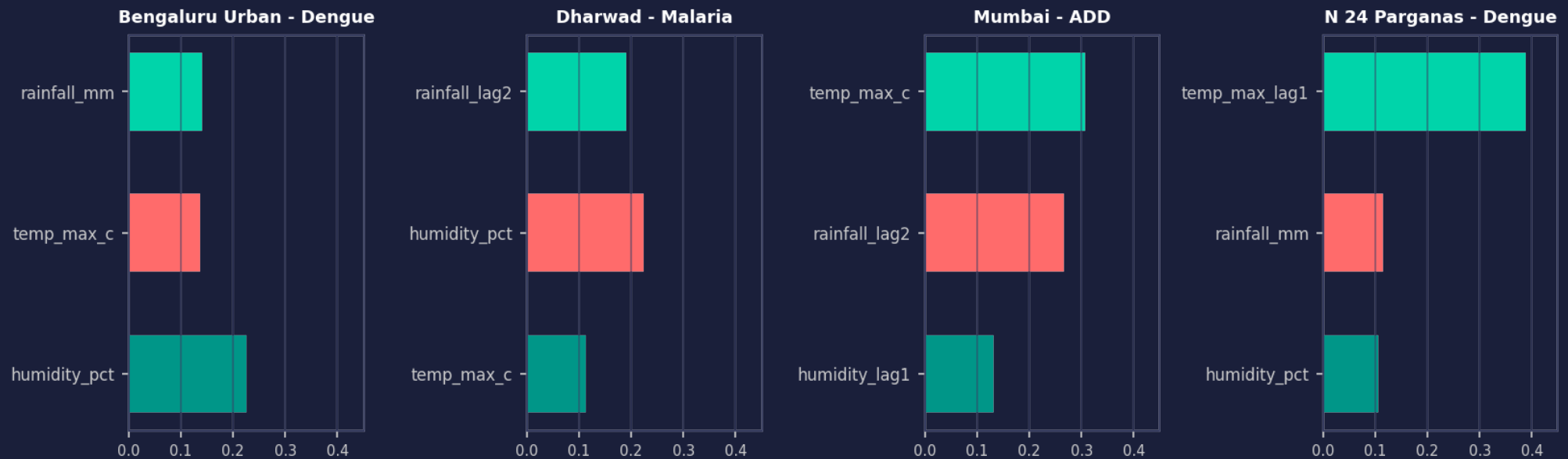


Predicted lead weeks: 6.5 weeks before the actual outbreak peak (2024-11-04, 89 cases)

Every prediction claim is auditable against [ml/results/backtest_results.json](#)

Explainability - SHAP Feature Importance

SHAP Feature Importance - Top 3 Climate Drivers per District-Disease



Key insight: temperature and rainfall lag features are consistently top drivers across districts - confirming the climate-disease linkage that powers every forecast.

Coverage - 9 Districts x 3 Diseases

District	State	Diseases Tracked
Bengaluru Urban	Karnataka	Dengue, Malaria, ADD
Dharwad	Karnataka	Dengue, Malaria, ADD
Mysuru	Karnataka	Dengue, Malaria, ADD
Mumbai	Maharashtra	Dengue, Malaria, ADD
Pune	Maharashtra	Dengue, Malaria, ADD
Nagpur	Maharashtra	Dengue, Malaria, ADD
Kolkata	West Bengal	Dengue, Malaria, ADD
Howrah	West Bengal	Dengue, Malaria, ADD
North 24 Parganas	West Bengal	Dengue, Malaria, ADD

54 trained models | 27 district-disease combinations x 2 model types (HGB baseline + XGB correction) | 0 failures

613,493 records across 11 Supabase PostgreSQL tables power the dashboard.

API Endpoints & Integration

Method	Endpoint	Description
GET	/health	Health check + DB connectivity
GET	/districts	List all 9 tracked districts
GET	/districts/{id}	Single district metadata
GET	/districts/{id}/forecast	8-week disease forecast
GET	/districts/{id}/history	Historical cases + climate
GET	/districts/{id}/risk	Risk tier by disease
GET	/methodology	Model pipeline + SHAP importance
GET	/backtest/{id}	Backtest event details
POST	/assistant/query	Grounded AI assistant query
40/40 integration tests pass Full audit trail (13/13 artifacts verified) SHA-256 data provenance manifest		
Static JSON fallbacks ensure the demo works even without network access.		

How EpiWatch Solves the Problem

1

1. DATA INGESTION

Weekly IDSP disease cases + NASA POWER climate + Census 2011 demographics

2

2. ETL PIPELINE

Join on district_id + week_start -> 11,232 rows x 12 features

3

3. FEATURE ENGINEERING

Seasonality, case lags (1/2/4 weeks), climate lags (2-wk rainfall, 1-day temp/humidity)

4

4. HGB BASELINE

HistGradientBoosting captures seasonal trend from case history

5

5. XGB CORRECTION

XGBoost learns how climate anomalies adjust the baseline via residuals

6

6. ENSEMBLE PREDICT

Final = HGB + XGB, 8-week forward forecast with confidence intervals

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7. RISK TIERING

Low -> Medium -> High -> Critical (cases per 100K), color-coded maps

8

8. AI ASSISTANT

Grounded Claude chat: every answer sourced from DB / SHAP / documents - refuses to hallucinate

RESULT: District-level 8-week outbreak forecasts weeks before peak | avg MAE 2.64 | zero training failures | fully auditable

Conclusion & Impact

EpiWatch Impact

- 9 Indian districts, 3 diseases covered
- 8-week advance warning before peak
- Avg MAE 2.64 across all combos
- 54 trained models, 0 failures
- SHA-256 auditable provenance
- Grounded AI with source citations
- Real-time risk maps for action

Future Directions

- Expand to all 738 Indian districts
- Add El Nino / La Nina signals
- Live NASA feed deployments
- Respiratory disease extension
- Deep learning seasonality
- Intervention optimization

Tech Stack

Next.js 15 / FastAPI 0.115 / Supabase PostgreSQL (613K records) / scikit-learn HistGBR + XGBoost / NASA POWER / Claude AI
Full audit trail: 40/40 integration tests · 13/13 artifacts verified · SHA-256 data provenance

EpiWatch - predicting outbreaks before they peak